

TECHNICAL MEMORANDUM

Employer: Trimac Engineering, Mechanical Engineer

To: Dr. Andrew Warkentin

From: Debaprashad Bhowmik

Cc:

WT#: WT 2

Date: September 13th, 2024

RE: Design and Development of Coupling Bolt and matching Hex Nut for 6MW Hydroelectric Power Station

1. Executive Summary
2. Introduction and Purpose
3. Theory and Analysis
4. Results and Discussion
5. Conclusion and Recommendations
6. References
7. Attachments: Summary of Data, Charts, Tables

Executive Summary

The coupling bolt is a vital mechanical component that connects two halves of a flanged shaft, ensuring proper torque transfer and maintaining alignment between the shafts. Misalignment can cause excessive vibration, increased bearing loads, and reduced operational efficiency. In the 6 MW hydroelectric power station in Nova Scotia, the coupling bolt links the generator shaft to the turbine shaft, handling torque loads of approximately 50,000 lb-ft. Trimac Engineering has undertaken the task of designing a new coupling bolt using AutoCAD for the power station. The bolt is specified as a stainless-steel rod with a diameter of 4¼ inches and a length of 9 inches, made from ASTM A193 Grade B6 material, known for its high tensile strength and corrosion resistance. It achieves a tensile strength of 150,000 psi to withstand operational stresses. Relief features are included to reduce stress concentrations, minimizing fatigue over 10 million load cycles. A smooth surface finish ($Ra \leq 3.2 \mu m$) is specified to reduce friction and wear between the bolt and the connected shafts, extending its operational life. Matching hex nuts are designed with a thread size of 2¼ inches - 8 UNC (Class 2B) and a height of 2.2 inches, made from zinc-plated steel per ASTM A194 Grade 2H standards. This design aligns with industry practices and ensures the hydroelectric station continues to operate reliably, with minimal risk of failure and reduced maintenance costs.

Introduction and Purpose

The design and development of a new coupling bolt for the 6MW hydroelectric power station and details the engineering considerations, theoretical analyses, and calculations that guided the design process, along with recommendations for future improvements are covered in this document. The power station, located in Nova Scotia, was originally commissioned in the 1950s. As part of a large-scale infrastructure, this power station plays a vital role in providing energy to the region. Given its age, regular maintenance is required to ensure continued operation and prevent unexpected failures. One of the most critical components needing replacement during maintenance cycles is the coupling bolt, which connects the generator shaft to the turbine shaft. This bolt serves as a key mechanical link that transfers torque between the shafts, ensuring smooth operation while maintaining precise alignment.

The power station undergoes scheduled shutdowns every 10-15 years to replace components that are intentionally designed to wear out. This proactive maintenance approach prevents catastrophic failures of more expensive machinery, such as the generator and turbine shafts, by allowing smaller, less expensive parts—like the coupling bolt—to fail first under stress. The bolt itself endures substantial mechanical loads.

The purpose of this project was to design a coupling bolt that not only meets the mechanical demands of the power station but also improves upon existing designs. The new bolt, designed using AutoCAD, incorporates several enhancements. These include the use of high-grade alloy steel to achieve a high tensile strength, providing a substantial margin over typical operating stresses. Relief features were carefully added to distribute stress evenly across the bolt, reducing the likelihood of fatigue failures over time. Additionally, the surface finish was optimized to minimize friction and wear, further extending the bolt's operational life. The initial design is shown in **Attachment 1**.

Theory and Analysis

The design of the Coupling bolt and Hex Nut was based on the following theory and analysis:

1. Coupling Bolt Material Selection

Selected Material: ASTM A193 Grade B6 (AISI 410 Stainless Steel)

Mechanical Properties

- Ultimate Tensile Strength ≥ 758 MPa (ASTM International, 2020a).
- Yield Strength ≥ 586 MPa (ASTM International, 2020a).
- Elongation $\geq 15\%$ in 50 mm (ASTM International, 2020a).
- Hardness Rockwell C35 maximum (ASTM International, 2020a).

Justification

Mechanical Strength: The calculated von Mises stress on the bolt is approximately 350 MPa. With a yield strength of 586 MPa, the safety factor is about 1.67, satisfying design requirements (Budynas & Nisbett, 2015).

Corrosion Resistance: AISI 410 stainless steel offers moderate corrosion resistance suitable for freshwater environments, preventing premature failure (Davis, 2000).

Manufacturability: The material allows for precise machining and achieving the required surface finish of $R_a \leq 3.2 \mu\text{m}$, essential for reducing stress concentrations and improving fatigue life (Kalpakjian & Schmid, 2014).

2. Hex Nut Material Selection

Selected Material: ASTM A194 Grade 2H Carbon Steel (Zinc-Plated)

Mechanical Properties

- **Proof Load:** 1,206 MPa (ASTM International, 2019).
- **Hardness:** Rockwell C24–38 (ASTM International, 2019).

Justification

Mechanical Strength: The proof load exceeds the preload applied to the bolt, ensuring the nut can withstand operational stresses without deformation (Shigley & Mischke, 2008).

Corrosion Resistance: Zinc plating provides initial corrosion protection by acting as a sacrificial anode, though its effectiveness may diminish over time in high-moisture environments (Jones, 1996).

Material Compatibility Concerns:

- **Galvanic Corrosion Risk:** Dissimilar metals in contact, such as stainless-steel bolts and zinc-plated carbon steel nuts, can lead to galvanic corrosion (Jones, 1996).

- **Mitigation Strategies:**

- **Use of Insulating Materials:** Introducing non-conductive washers to electrically isolate the bolt and nut (Davis, 2000).
- **Alternative Material:** Selecting nuts made from compatible stainless steel, such as ASTM A194 Grade 8 (AISI 304/316), to match the bolt material and eliminate galvanic potential (ASTM International, 2019).

3. **Surface Finish:** The surface finish of the coupling bolt is a critical factor influencing its mechanical performance, particularly in terms of fatigue life and wear resistance. A smoother surface reduces micro-level stress concentrations, thereby enhancing fatigue strength and decreasing the likelihood of crack initiation (Budynas & Nisbett, 2015).

Specified Surface Roughness: The coupling bolt is specified to have a surface roughness (Ra) of $\leq 3.2 \mu\text{m}$. This level of finish corresponds to standard machining processes, achievable through fine turning or milling operations (Kalpakjian & Schmid, 2014). It offers an optimal balance between manufacturing feasibility and improved mechanical properties.

Impact on Fatigue Strength: Surface roughness affects the material's endurance limit. The surface finish factor adjusts the baseline endurance limit to account for surface conditions (Budynas & Nisbett, 2015):

$$\sigma_e = \sigma'_e \times k_s = 0.5\sigma_u \times 0.85$$

Assuming an ultimate tensile strength of 758 MPa:

$$\sigma_e = 0.5 \times 758 \text{ MPa} \times 0.85 = 322 \text{ MPa}$$

Using the Modified Goodman criterion:

$$\frac{\sigma_a}{\sigma_e} + \frac{\sigma_m}{\sigma_u} = \frac{46.5 \text{ MPa}}{322 \text{ MPa}} + \frac{348.7 \text{ MPa}}{758 \text{ MPa}} = 0.144 + 0.460 = 0.604 < 1$$

This calculation indicates the bolt meets fatigue life requirements with the specified surface finish (Stephens et al., 2000).

Effect on Wear and Friction: A smoother surface reduces friction and wear by minimizing asperity interactions between contact surfaces (Williams, 2005).

According to Rabinowicz's (1995) wear equation, decreasing surface roughness lowers the wear coefficient, reducing material loss over time.

Manufacturing Considerations: Achieving $R_a \leq 3.2 \mu\text{m}$ requires precise control of machining parameters such as cutting speed, feed rate, and tool selection (Kalpakjian & Schmid, 2014). Quality assurance through surface profilometry ensures compliance with specifications (Whitehouse, 2011).

4. Fatigue Life Estimation

The bolt must withstand cyclic stresses over its operational life.

Total Cycles (N):

Assuming continuous operation over 15 years:

- Hours per year: $365 \times 24 = 8,760$ hours
- Total hours: $8,760 \text{ hours/year} \times 15 \text{ years} = 131,400$ hours
- Rotational speed: $N = 180 \text{ rpm}$
- Total cycles:

$$N = 131,400 \text{ hours} \times 60 \text{ min/hour} \times 180 \text{ rev/min} = 1,418,280,000 \text{ cycles}$$

The fatigue life estimation shows that the coupling bolt, made from ASTM A193 Grade B6 stainless steel, is designed to endure approximately 1.42 billion cycles over its operational life. Based on the stress analysis and fatigue strength of the material, the bolt will reliably perform under cyclic loading, meeting the operational demands of the hydroelectric power station.

Results and Discussion

The coupling bolt, designed using ASTM A193 Grade B6 (AISI 410 stainless steel), is expected to handle torque loads of approximately 50,000 lb-ft while maintaining a high safety margin. The mechanical properties of the selected material, including an ultimate tensile strength of 758 MPa and a yield strength of 586 MPa, ensure that the bolt will not fail under operational stresses. The calculated von Mises stress of 350 MPa results in a safety factor of 1.67, indicating that the design meets the station's mechanical requirements.

Furthermore, fatigue analysis demonstrates that the coupling bolt can endure over 1.42 billion cycles over its 15-year service life, assuming a continuous operational speed of 180 rpm. Relief features incorporated into the design reduce stress concentrations, extending the bolt's fatigue life. These features, combined with a surface finish of $Ra \leq 3.2 \mu\text{m}$, ensure the bolt will resist fatigue and wear, meeting the expected operational life without premature failure.

The hex nut was designed using ASTM A194 Grade 2H carbon steel with a zinc-plated coating for corrosion protection. The proof load of 1,206 MPa ensures that the nut can handle the preload stresses without deforming. However, zinc plating presents a potential risk of galvanic corrosion when paired with the stainless-steel coupling bolt. This corrosion risk could reduce the overall service life of the assembly if not addressed.

To mitigate this risk, the use of non-conductive washers or alternative stainless-steel nuts (ASTM A194 Grade 8) is recommended. These measures will prevent galvanic coupling and ensure long-term performance under varying environmental conditions.

The surface finish of the coupling bolt was specified at $Ra \leq 3.2 \mu\text{m}$, reducing friction and wear between the bolt and connected shafts. A smoother surface minimizes micro-asperities, which reduces the risk of fatigue crack initiation. The specified surface finish, combined with the bolt's material properties, ensures that the fatigue strength remains within acceptable limits throughout its operational life.

The use of stainless steel for the coupling bolt provides moderate resistance to corrosion in freshwater environments. The zinc-plated coating on the hex nut offers sacrificial protection, though its long-term effectiveness in high-moisture environments may diminish. Routine maintenance and inspections are recommended to ensure corrosion protection is maintained over time. Consideration should be given to the potential upgrade to more corrosion-resistant materials for extended operational life. A flowchart illustrating the entire process can be found in **Attachment 2**.

Conclusion and Recommendations

The coupling bolt, designed using ASTM A193 Grade B6 stainless steel, meets the operational requirements of the 6 MW hydroelectric power station. Its mechanical properties ensure that it can handle the required torque loads and endure over 1.42 billion cycles, aided by relief features and a surface finish of $R_a \leq 3.2 \mu\text{m}$ to reduce wear and fatigue. However, the risk of galvanic corrosion between the stainless-steel bolt and the zinc-plated carbon steel hex nut remains a concern. While the current hex nut design satisfies mechanical stress requirements, the corrosion risk should be addressed for long-term reliability. Overall, the design of the coupling bolt and hex nut provides a reliable solution for the hydroelectric power station, with opportunities for further enhancement in material selection to improve corrosion resistance. The professionally completed drawing, adhering to industry standards, is provided in **Attachment 3**. To ensure long-term reliability and performance, the following recommendations are proposed:

- Proceed with ASTM A193 Grade B6 for the coupling bolt due to its high mechanical strength and corrosion resistance.
- Implement proper hardening and tempering to achieve the required tensile and fatigue properties.
- Maintain a surface roughness of $R_a \leq 3.2 \mu\text{m}$ to reduce friction and enhance fatigue life.
- Consider switching to ASTM A194 Grade 8 stainless steel to prevent galvanic corrosion.
- If carbon steel nuts are retained, ensure robust zinc plating and consider additional coatings to suit environmental conditions.
- Conduct regular inspections and maintenance to monitor corrosion and wear in the assembly.
- Implement an Inspection and Test Plan (ITP), similar to **Attachment 4**, to thoroughly inspect the coupling bolt and hex nut assembly at critical stages of the manufacturing and operational process.

References

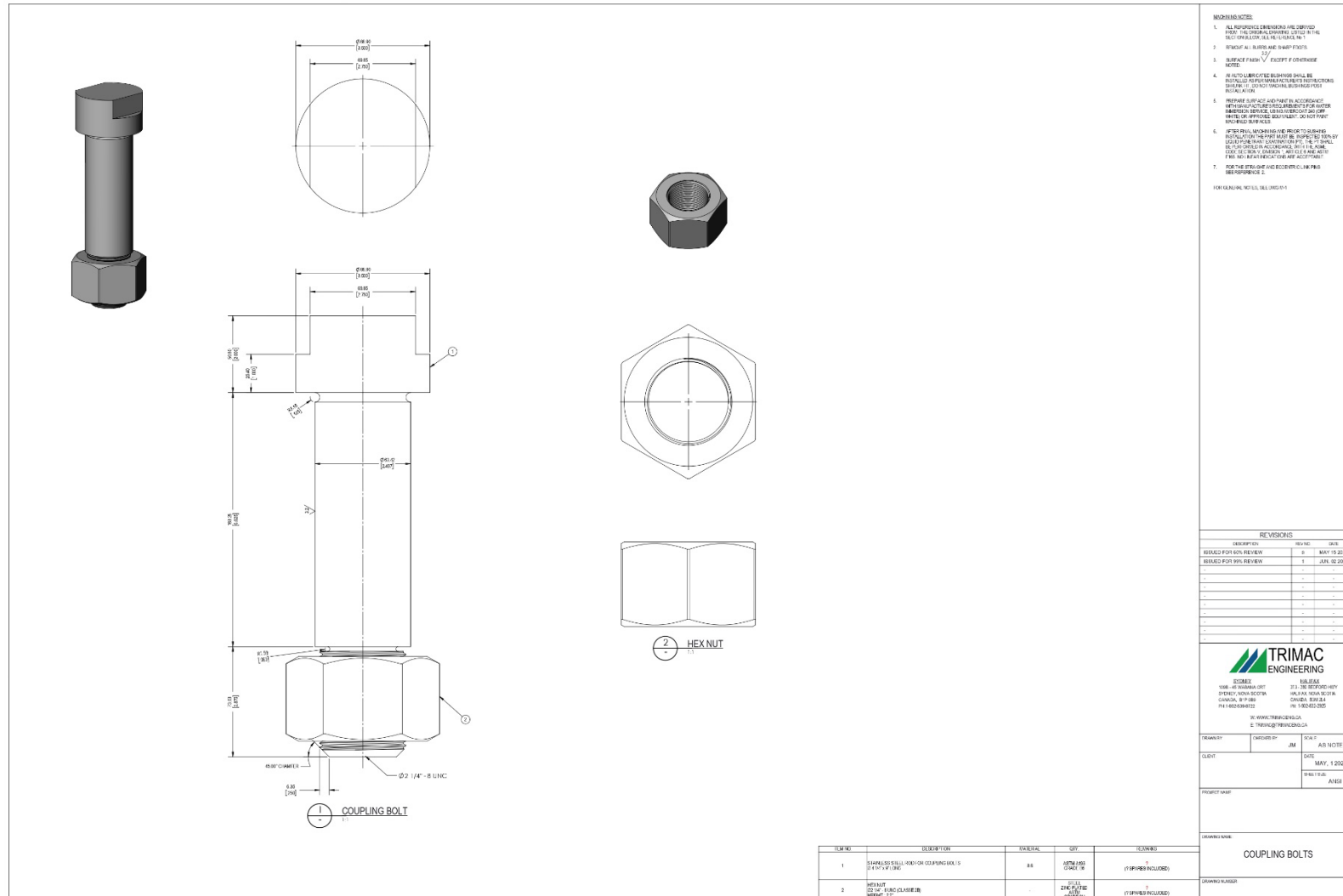
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Attachment 2



Note. This Flowchart is produced by me.

Attachment 3



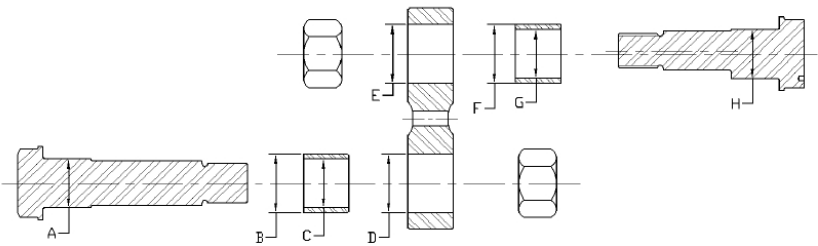
Note. This drawing with professional title block is produced by me.

Attachment 4

PLANT NAME / UNIT: HYD Cowie 12 Overhaul
REPORT DATE: _____
COMPLETED BY: _____

INSPECTION AND TEST PLAN - HYDRO MAINTENANCE SHOP
LINKS, BUSHINGS PINS

DOC# QR-T-XX



MEASUREMENTS - COUPLING BOLT, SHAFT			
SPEC	LOCATION		A
	Max (in)		1.8755
	Min (in)		1.8745
		0°	90°
SET 1			
SET 2			
SET 3			
SET 4			
SET 5			
SET 6			
SET 7			
SET 8			
SET 9			
SET 10			
SET 11			
SET 12			
SET 13			
SET 14			

Note. This ITP sheet is produced by me.